



EGI VO

SERVICE LEVEL AGREEMENT

Customer	MoBRAIN/vo:enmr.eu/vo.neugrid.eu
Provider	EGI.eu
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DOCUMENT LOG

<i>Issue</i>	<i>Date</i>	<i>Comment</i>	<i>Author</i>
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TERMINOLOGY

The EGI glossary of terms is available at: <https://wiki.egi.eu/wiki/Glossary>

For the purpose of this Agreement, the following terms and definitions apply. The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in RFC 2119.

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The present Service Level Agreement (“the Agreement”) is made between **EGI.eu (the Provider)** and **MoBRAIN/vo:enmr.eu/vo.neugrid.eu (the Customer)** to define the provision and support of the provided services as described hereafter. Representatives and contact information are defined in Section 6.

The main objective of MoBrain is to lower barriers for scientists to access modern e-Science solutions from micro to macro scales. By building on grid- and cloud-based infrastructures and on the existing expertise available within WeNMR¹ and N4U² and by integrating molecular structural biology and medical imaging services and data, MoBrain will kick-start the development of a larger, integrated, global science virtual research environment for life and brain scientists worldwide. The projects defined in MoBrain are geared toward facilitating this overall objective, each with specific objectives to reinforce existing services, develop new solutions and pave the path to global competence centre and virtual research environment for transnational research from molecular to brain.

The Customer is a consortium represented by **the Faculty of Science – Chemistry, Utrecht University**.

This Agreement is valid from **01/01/2016** to **01/01/2017**.

The Agreement was discussed and approved by the Customer and the Provider on **16/02/2016**.

1 The Services

The Services are defined by the following properties:

High-Throughput Compute (category: Compute) and File Storage (category: Storage)

A High-Throughput Compute allows running computational tasks on high quality IT resources, accessible via a uniform/standard interface and supporting authentication/authorisation based on a membership within a virtual organisation. HTC Compute service is federated from EGI Federation providers offering seamless access to computing capabilities with integrated monitoring and accounting.

File storage is provided remotely on different Resource Providers with different storage standard interfaces that are transparently available with the possibility of replication.

- Resource Centre: **INFN-PADOVA** (Country: Italy)
 - High-Throughput Compute
 - Opportunistic computing time [HEPSPEC-hours]: 7 Millions
 - Max job duration [hours]: 72
 - Min local storage [GB] (scratch space for each core used by the job): 1 GB

¹ www.wenmr.eu

² neugrid4you.eu

- Min physical memory per core [GB]: 2 GB
 - Other technical requirements: nodes with 24 AMD CPU-cores
 - Middleware: gLite CREAM-CE
 - Duration: 01/01/2016 – 01/01/2017
 - File Storage
 - Opportunistic storage capacity [TB]: 9 TB
 - Duration: 01/01/2016 – 01/01/2017
 - Supported VOs: enmr.eu
 - Access mode offer: LSF fair-share for HTC
- Resource Centre: **RAL-LCG2** (Country: United Kingdom)
 - High-Throughput Compute
 - Opportunistic computing time [HEPSPEC-hours]: 5 Millions
 - Max job duration [hours]: 72
 - Min local storage [GB] (scratch space for each core used by the job): 50 GB
 - Min physical memory per core [GB]: 2 GB
 - Other technical requirements: 130 Sandy Bridge cores configured to support hyper-threading.
 - Middleware: CVMFS Stratum 0 or 1. SRM gridftp/xrootd or trial S3 interface to CEPH
 - Duration: 01/01/2016 – 01/01/2017
 - File storage
 - Opportunistic storage capacity [GB]: 50 GB
 - Duration: 01/01/2016 – 01/01/2017
 - Supported VOs: enmr.eu
 - Access mode offer: Opportunistic. Resources are not exclusively allocated but subjected to local availability
- Resource Centre: **TW-NCHC** (Country: Taiwan)
 - High-Throughput Compute
 - Opportunistic computing time [HEPSPEC-hours]: 2 Millions
 - Max job duration [hours]: 8
 - Min local storage [GB] (scratch space for each core used by the job): 5 GB
 - Min physical memory per core [GB]: 2 GB
 - Middleware: gLite EMI-3, CVMFS
 - Duration: 01/01/2016 – 01/01/2017
 - File storage
 - Opportunistic storage capacity [TB]: 50 TB
 - Duration: 01/01/2016 – 01/01/2017
 - Supported VOs: enmr.eu
 - Access mode offer: Opportunistic
- Resource Centre: **SURFsara** (Country: The Netherlands)
 - High-Throughput Compute
 - Opportunistic computing time [HEPSPEC-hours]: 20 Millions
 - Max job duration [hours]: 36 hours (medium queue), 72-96 hours (long queue)
 - Min local storage [GB] (scratch space for each core used by the job): 75 - 200 GB / core

- Min physical memory per core [GB]: 4 - 8 GB
 - Middleware: UMD3 (gLite, EMI)
 - Duration: 01/01/2016 – 01/01/2017
 - File Storage
 - Opportunistic storage capacity [GB]: 50 GB
 - Duration: 01/01/2016 – 01/01/2017
 - Supported VOs: enmr.eu
 - Access mode offer: Opportunistic via Grid
- Resource Centre: **NCG-INGRID-PT** (Country: Portugal)
 - High-Throughput Compute
 - Opportunistic computing time [HEPSPEC-hours]: 1 Million per year
 - Max job duration [hours]: 52
 - Min local storage [GB] (scratch space for each core used by the job): 10 GB
 - Min physical memory per core [GB]: 2 GB
 - Middleware: UMD3 (gLite, EMI)
 - Duration: 01/01/2016 – 01/01/2017
 - Supported VOs: enmr.eu
 - Access mode offer: Opportunistic
- Resource Centre: **NIKHEF** (Country: The Netherlands)
 - High-Throughput Compute
 - Opportunistic computing time [HEPSPEC-hours]: 20 Millions
 - Max job duration [hours]: 36 hours (medium queue), 72-96 hours (long queue)
 - Min local storage [GB] (scratch space for each core used by the job): 60 - 200 GB / core
 - Min physical memory per core [GB]: 3 - 8 GB
 - Middleware: UMD3 (gLite, EMI)
 - Duration: 01/01/2016 – 01/01/2017
 - File Storage
 - Opportunistic storage capacity [GB]: 50 GB
 - Duration: 01/01/2016 – 01/01/2017
 - Access mode offer: Opportunistic via Grid interface
 - Supported VOs: enmr.eu
- Resource Centre: **CESNET-MetaCloud** (Country: Czech Republic)
 - Cloud Compute
 - Number of Virtual CPU cores: up to 60
 - Memory: up to 360 GB
 - Scratch/ephemeral storage: 4TB/2TB
 - Public IP addresses: yes (up to 10, more available upon request)
 - Access type: pledged
 - Duration: 01/01/2016 – 01/01/2017
 - Supported VOs: enmr.eu

The Services are supported by additional services (category: Software and Service Platform):

- Accounting³
- Service Monitoring⁴ (operational only)

Note: Please note that following services are not provided by EGI.eu:

- Monitoring of the VOs: vo:enmr.eu and vo.neugrid.eu
- Monitoring of services provided by the Customer on agreed resources

2 Service hours and exceptions

The Services operate during the following hours: twenty-four (24) hours a day, seven (7) days a week, three hundred sixty-five (365) days a year.

The following exceptions apply:

- Planned maintenance windows or service interruptions (“scheduled downtimes”⁵) will be notified via e-mail in a timely manner i.e. 24 hours before the start of the outage⁶.
- Downtime periods exceeding 24 hours need justification.

3 Support

Support is provided via EGI Service Desk⁷. Access requires a valid X.509 or the login via a EGI SSO account⁸. Support is available between:

- Monday to Friday.
- From 9:00 to 17:00 in the time zone of the relevant Resource Centres.

Service times always apply with the exception of public holidays in the country of the supporting Resource Centres.

3.1 Incident handling

Incidents will be handled according to the Quality of Support level that is estimated according to the impact of the outage or service quality degradation.

³ <http://accounting.egi.eu/>

⁴ <http://argo.egi.eu/>

⁵ https://wiki.egi.eu/wiki/GOCDB/Input_System_User_Documentation#Downtimes

⁶ <http://goc.egi.eu/>

⁷ <http://helpdesk.egi.eu/>

⁸ <https://www.egi.eu/sso/>

The Quality of Support in this Agreement has level: **Medium**⁹.

Incident priority ¹⁰	Response time
Less urgent	5 working days
Urgent	5 working days
Very Urgent	1 working day
Top Priority	1 working day

Table 1. Response times to incidents according to the incident priority of “Medium” services

3.2 Service requests

In addition to resolving incidents, standard service requests (e.g. change requests, information requests, documentation) will be fulfilled through the defined support channels in the same way as incidents. Service requests are classified as “Less urgent”.

4 Service level targets

Monthly Availability

- Defined as the ability of a service or service component to fulfil its intended function at a specific time or over a calendar month.
- Minimum (as a percentage per month):
 - INFN-PADOVA: 85%
 - RAL-LCG2 : 85%
 - TW-NCHC : 90%
 - NIKHEF : 85%
 - SURFsara: 85%
 - NCG-INGRID-PT: 85%
 - CESNET-MetaCloud: 85%

Monthly Reliability

- Defined as the ability of a service or service component to fulfil its intended function at a specific time or over a calendar month, excluding scheduled maintenance periods.
- Minimum (as a percentage per month):
 - INFN-PADOVA: 90%
 - RAL-LCG2 : 90%
 - TW-NCHC : 90%

⁹ https://wiki.egi.eu/wiki/FAQ_GGUS-QoS-Levels

¹⁰ https://wiki.egi.eu/wiki/FAQ_GGUS-Ticket-Priority

- NIKHEF: 90%
- SURFsara: 90%
- NCG-INGRID-PT: 90%
- CESNET-MetaCloud: 90%

Quality of Support level

- Medium (Section 3)

5 Limitations and constraints

The Services provisioning is subject to the following limitations and constraints.

- Support is provided in English.
- Availability and Reliability calculations are based on the Service Monitoring operational results.
- Failures in vo:enmr.eu and vo.neugrid.eu monitoring are not considered as the Agreement violations.
- Downtimes needed to ensure the security of the Service issues are not considered Agreement violations.
- Failures of resource provider not being part of EGI production infrastructure are not considered as Agreement violations.
- Force Majeure. A party shall not be liable for any failure of or delay in the performance of this Agreement for the period that such failure or delay is due to causes beyond its reasonable control. Means any
 - fire, flood, earthquake or natural phenomena,
 - war, embargo, riot, civil disorder, rebellion, revolution
 which is beyond the Provider's control, or any other causes beyond the Provider's control
- Failures of the CVMSF system, outside the site providing the service, but affecting the service are not considered as the Agreement violation.

6 Communication, reporting and escalation

6.1 General communication

The following contacts will be generally used for communications related to the service in the scope of this Agreement.

Customer contact for the Provider	Alexandre Bonvin a.m.j.j.bonvin@uu.nl
Provider contact for the Customer	Małgorzata Krakowian sla@mailman.egi.eu SLA Coordinator at EGI.eu
Service Support contact	See Section 3

6.2 Regular reporting

As part of the fulfilment of this Agreement and provisioning of the service, the following reports will be provided:

Report title	Contents	Frequency	Delivery
Services Performance Report	The document provides the overall assessment of service performance (per month) and SLA target performance achieved during last 6 months	Every six months	Email to the Customer

6.3 Violations

The Provider commits to inform the Customer, if this Agreement is violated or violation is anticipated. The following rules are agreed for communication in the event of violation:

- In case of violations of the Services targets for two consecutive months, the Provider will provide justifications and a plan for Services enhancement to the Customer. The Provider will produce a status report and a Service enhancement plan for the improvement of the Services within one month from the date of the first notification.
- The Customer will notify the supporting Resource Centres in case of suspected violation via the EGI Service Desk (Section 3). The case will be analysed to identify the cause and verify the violation.

6.4 Escalation and complaints

For escalation and complaints, the Provider contact (section 6.1) point shall be used, and the following rules apply.

- In case of repeated violation of the Services targets for four consecutive months, a review of the Agreement and of the Services enhancement plan will take place involving the parties of the Agreement.
- Complaints or concerns about the Services provided should be directed to the Provider contact who will promptly address these concerns. Should the Customer still feel dissatisfied, about either the result of the response or the behaviour of the supporting Resource Centres, EGI.eu Director director@egi.eu should be informed.

7 Information security and data protection

The following rules for information security and data protection related to the Service apply.

- Assertion of absolute security in IT systems is impossible. The Provider is making every effort to maximize security level of users' data and minimize possible harm in the event of an incident.
- The Provider will define and abide by an information security and data protection policy related to the service being provided.
- The parties of the Agreement will meet all requirements of any relevant policy or procedure of the Provider¹¹ and will comply with the applicable national legislations.

8 Responsibilities

8.1 Of the Provider

Additional responsibilities of the Provider are as follows.

- The Provider adheres to all applicable operational and security policies and procedures¹² and to other policy documents referenced therein.
- The Provider allows monitoring of the Service in order to measure the fulfilment of the agreed service level targets.
- The Provider retains the right to introduce changes in how the Service is provided, in which case the Provider will promptly inform the Customer and update the Agreement accordingly.

8.2 Of the Customer

The responsibilities of the Customer are:

¹¹ https://www.egi.eu/about/policy/policies_procedures.html

¹² https://www.egi.eu/about/policy/policies_procedures.html

- The Customer must not share access credentials with anyone else.
- The data stored in the system by the Customer must not cause any legal violation due to the content type (such as copyright infringement, dual use, illegal material).
- The use must be consistent with the Acceptable Use Policy¹³ of the Service.
- The Customer will notify the Provider about foreseen under- (in case of guaranteed allocation) or over-estimated actual amount of the Service used to update of the Agreement to ensure optimal usage of the Service.
- The Customer should have registered one or more Virtual Organizations (VOs) to define the group of users entitled to access the Service. Information about the VO will be regularly updated in EGI Operations Portal¹⁴.
- The Customer must request EGI Service Desk support¹⁵ to enable assigning tickets with appropriate VO name.
- When applicable, the Customer is responsible of ensuring that the Virtual Machine images endorsed and listed in the AppDB¹⁶ VO image list are properly maintained and updated.
- The Customer commits to acknowledge EGI in the scientific publications benefiting from the Service: "The FP7 WeNMR (project# 261572) and H2020 West-Life (project# 675858) European e-Infrastructure projects are acknowledged for the use of their web portals, which make use of the EGI infrastructure with the dedicated support of CESNET-MetaCloud, INFN-PADOVA, NCG-INGRID-PT, RAL-LCG2, TW-NCHC, SURFsara and NIKHEF, and the additional support of the national GRID Initiatives of Belgium, France, Italy, Germany, the Netherlands, Poland, Portugal, Spain, UK, South Africa, Malaysia, Taiwan and the US Open Science Grid."
- The Customer will provide during Agreement review (yearly) list of scientific publications benefiting from the Service.

9 Review, extensions and termination

The Services performance will be reviewed against the defined Service level targets according to Section 4. The Agreement will be annually reviewed until expiration.

If the Customer wishes to extend the duration after the Agreement termination date, an extension will be negotiated with the Provider.

The Provider retains the right to introduce changes in the Service, in which case the Customer retains the right of terminating the Agreement.

¹³ <https://documents.egi.eu/document/74>

¹⁴ <http://operations-portal.egi.eu/>

¹⁵ https://wiki.egi.eu/wiki/FAQ_GGUS-New-Support-Unit

¹⁶ <https://appdb.egi.eu/>

The Agreement can be terminated at any time upon agreement of the parties. Amendments, comments and suggestions must be addressed to the Provider and the Customer according to Section 6.